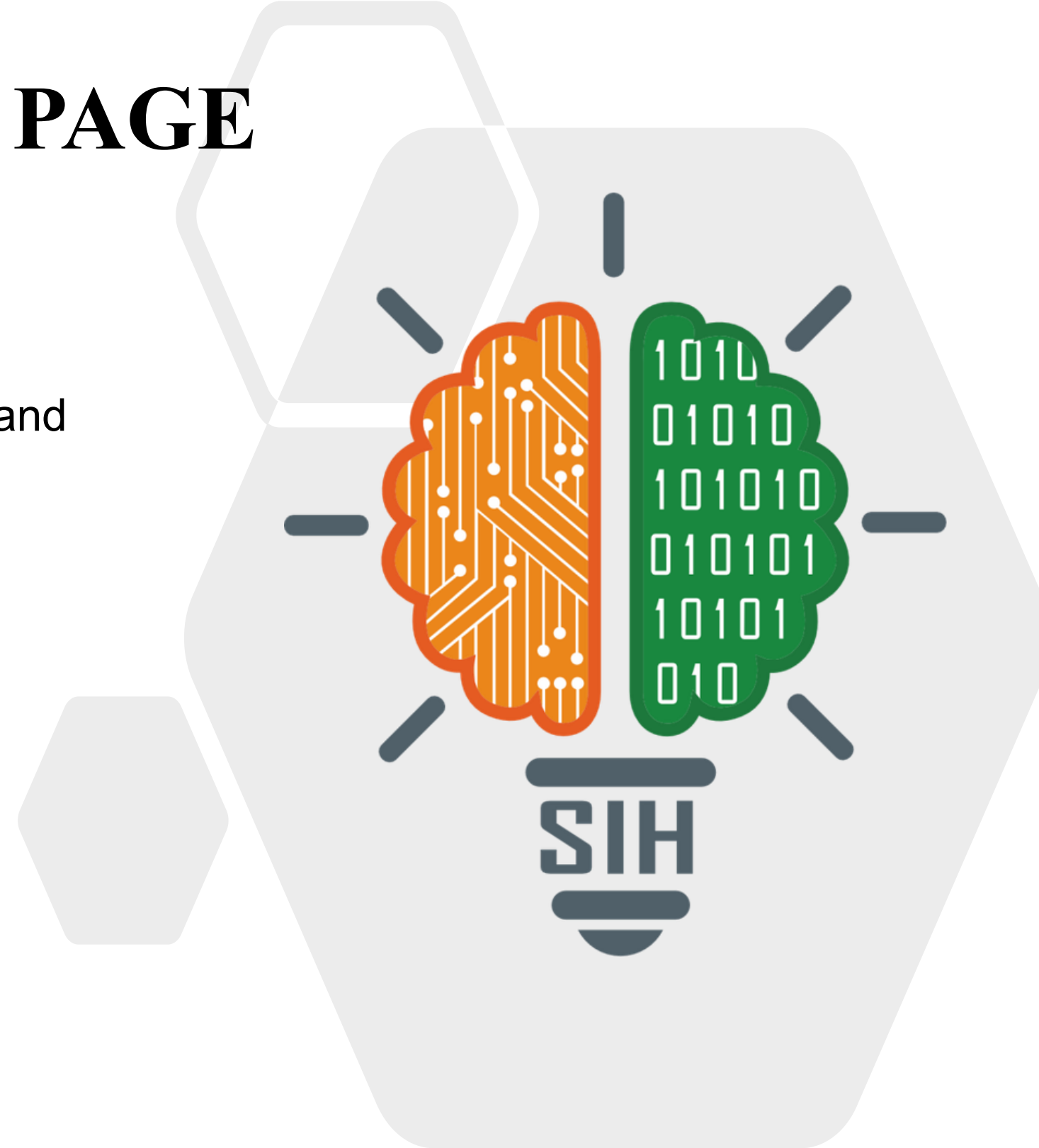


TITLE PAGE

- **Problem Statement ID** – SIH26037
- **Problem Statement Title** – Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads
- **Theme** – Smart Vehicles
- **PS Category** – Software
- **Team ID** – 161638
- **Team Name** – Ctrl Alt Elite



Adaptive Path Planning & Collision Avoidance for Unstructured Indian Roads



PROPOSED SOLUTION

Proposed Solution (Describe your Idea/Solution/Prototype) · Detailed explanation of the proposed solution

PathSense is an adaptive path-planning and collision-avoidance system for autonomous vehicles on unstructured Indian roads. It reasons about **mixed traffic** — pedestrians, motorcycles, auto-rickshaws, cars, buses, trucks and animals — on roads with **missing or unclear lane markings**, informal merges, sudden direction changes, wrong-way movement and temporary obstacles.



concept illustration



PROBLEMS IT ADDRESSES

How it addresses the problem

01



Unmarked / Unstructured Roads

Plans in the drivable corridor, not painted lanes

02



Mixed Traffic

Cars, buses, trucks, 2W, autos, people, animals together

03



Irregular Road-User Behaviour

Reads intent from short-term motion history

04



Pedestrian & Cyclist Interaction

Acts on people entering the path, not the footpath

05



Wrong-Way / Informal Movement

Flags movement that enters the ego path

06



Animal & Temporary Obstacles

Plans around cattle, carts and work zones

07



Sudden Direction Changes

Re-evaluates every road user every frame

08



Dynamic Path Replanning

Keeps choosing a new safe path as the scene changes



DIFFERENTIATION & KEY VALUE PROPOSITION

Innovation and uniqueness of the solution

01

Ego-Path-Centric Risk Assessment

Judges the threat to the vehicle's own path — not mere proximity.

02

Indian Mixed-Traffic Awareness

Built for autos, two-wheelers, pedestrians and animals.

03

Short-Term Motion Understanding

Tracks each road user's motion to anticipate conflicts.

04

Adaptive Trajectory Generation

Samples kinematically feasible candidate paths every frame.

05

Lane-Independent Planning

Works where lane markings are faded, partial or absent.

06

Safety-Oriented Decision Making

GO → SLOW → STEER → BRAKE → NO SAFE PATH, each with a reason.

07

Dynamic Replanning

Re-plans continuously as traffic and obstacles change.

08

Scalable Multi-Sensor Roadmap

Camera-first today; radar, LiDAR and vehicle-state inputs later.

CORE
IDEA

Perceive
road scene

Understand
road users

Estimate
short-term movement

Identify
ego-path threats

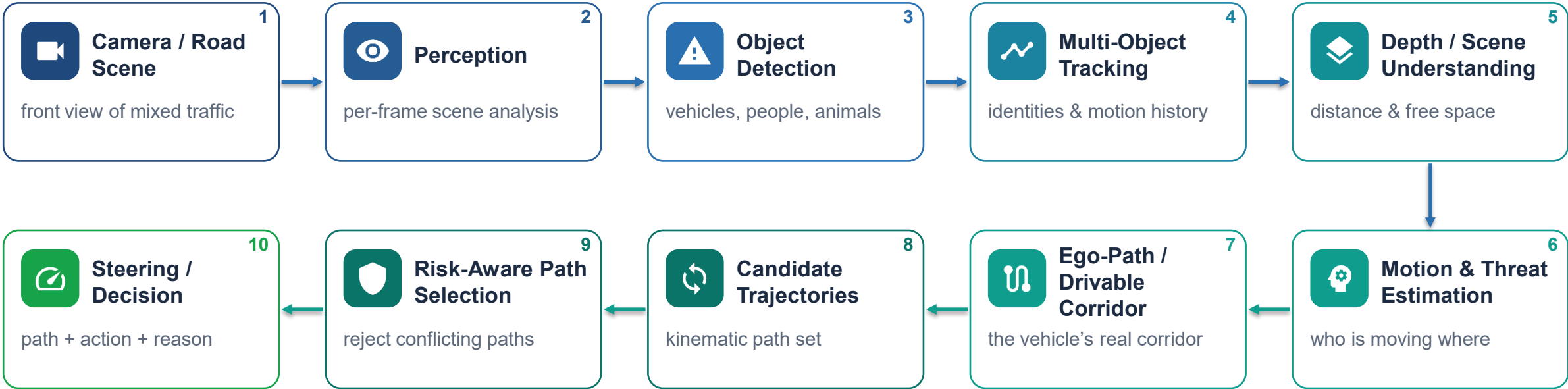
Generate
candidate trajectories

Select
a safe trajectory

Re-plan
continuously

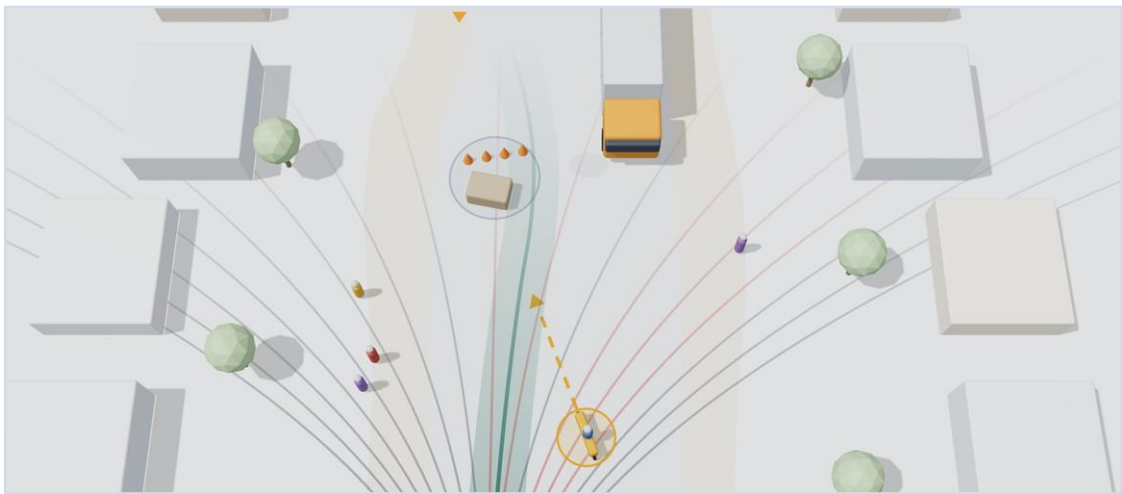
METHODOLOGY & ARCHITECTURE
Methodology and process for implementation (Flow Charts/Images/ working prototype)

SENSE & UNDERSTAND → row 1 **PLAN & DECIDE** → row 2 (right to left)



DECISION LAYER — escalates with the threat to the ego path

- GO** path clear, or traffic stays on its own side
- SLOW DOWN** a road user is approaching the path
- STEER** a safer offset trajectory exists
- BRAKE** a road user is entering / in the path
- NO SAFE PATH** every candidate path is blocked → stop



Candidate trajectories: conflicting paths (red) rejected, safe path (teal) kept — concept illustration

TECH STACK
Technologies to be used (e.g. programming languages, frameworks, hardware)

 Python core language	 OpenCV video & vision
 YOLO object detection	 PyTorch deep-learning runtime
 ByteTrack multi-object tracking	 Depth Anything V2 monocular depth
 NumPy planning & geometry	 CUDA GPU acceleration
 Flask demo web server	 HTML / CSS / JS review interface



FROM CAMERA VIEW TO DECISION

Actual output frames from the processed demo video — New BEL Road, Bengaluru (consecutive moments)

1 GO STRAIGHT



Oncoming car stays on its own side — ego path clear, keep going.

2 SLOW DOWN



Oncoming vehicle starts entering the ego path — threat approaching.

3 BRAKE



Pedestrian in the ego path — collision risk, brake.



BEFORE vs AFTER

Same instant — raw camera frame vs. PathSense output (Nandidurga Road, Bengaluru)



BEFORE — raw camera view



AFTER — PathSense: road users tracked, path clear → GO

CURRENT DEVELOPMENT

Camera-based proof-of-concept demonstrating perception, threat assessment and adaptive path decisions on mixed Indian-road scenes.

- Detects and tracks cars, buses, two-wheelers, auto-rickshaws, pedestrians and animals
- Separates traffic on its own side from threats entering the ego path
- Plans around obstacles with candidate trajectories
- Explains every decision and replays key moments



Source code

github.com/MalyalaKarthik66/pathsens-e-prototype

Frames are unedited pipeline output, cropped to the camera view. Source footage: L. Shyamal, Wikimedia Commons (CC0). Live demo shown separately.



PoC PROCESS — KEY STEPS

Define Problem & Objective · Set Success Criteria · Plan Scope & Resources · Build Prototype / Model · Test & Validate · Decide Next Steps

1



PROBLEM

Unstructured Indian roads make lane-based autonomous planning difficult.

2



OBJECTIVE

Understand dynamic road interactions and generate safe adaptive paths.

3



PROTOTYPE

Camera-based perception, tracking, motion understanding and path planning.

4



TEST & VALIDATE

Scenario-based evaluation across diverse Indian-road conditions.

5



SUCCESS CRITERIA

Appropriate path selection and risk response under changing traffic conditions.

6



NEXT STEPS

Sensor fusion, simulation / closed-loop validation and controlled vehicle integration.



SCOPE & RESOURCES

Plan Scope & Resources

IN SCOPE — proof of concept

- Camera-based perception of mixed Indian traffic
- Multi-object tracking and short-term motion cues
- Ego-path threat assessment and candidate-path planning
- Explainable GO / SLOW / STEER / BRAKE decisions on recorded road video

LATER — pilot / production

- Vehicle actuation and closed-loop control
- Radar / LiDAR and vehicle-state fusion
- Functional-safety engineering (ISO 26262 / SOTIF)
- Controlled field trials at scale

RESOURCES

- Open pretrained models (YOLO, ByteTrack, Depth Anything V2)
- Openly licensed Indian road-scene videos
- Python / PyTorch stack on a laptop-class GPU
- Simulation tools for the pilot stage



FEASIBILITY

Analysis of the feasibility of the idea

Technology: Established computer-vision, object-tracking, depth-estimation and trajectory-planning techniques provide a practical development base.

Implementation: Modular perception → prediction → planning architecture enables incremental development and validation.

Integration: Can progressively incorporate vehicle-state, radar and LiDAR inputs.

Validation: Scenario-based testing progresses from recorded road scenes to simulation and controlled field trials.



RISKS → MITIGATION

Potential challenges and risks · Strategies for overcoming these challenges

Unstructured road geometry

→ Drivable-corridor and ego-path estimation

Irregular road-user behaviour

→ Temporal tracking and short-term motion reasoning

Perception uncertainty

→ Confidence-aware decisions and future sensor fusion

Real-world validation complexity

→ Progressive scenario-based simulation and controlled testing



DATA-DRIVEN / OPERATIONAL VIABILITY

Data Foundation: road video, traffic observations, map / context information and future multi-sensor inputs.

Operational Value: adaptive navigation in mixed and unstructured road environments.

Scalability: prototype → pilot → production deployment pathway.

WHY IT MATTERS — India, 2022

4,61,312 road accidents **1,68,491** deaths

Source: MoRTH, Road Accidents in India 2022



PROTOTYPE → PILOT → PRODUCTION

Development roadmap

PROTOTYPE

Camera-based proof-of-concept for perception, tracking and adaptive path planning.

PILOT

Broader scenario validation, sensor expansion, simulation and controlled road trials.

PRODUCTION

Vehicle-grade multi-sensor perception, closed-loop validation and automotive safety engineering.



POTENTIAL COLLABORATIONS

Industry · academia · government



Automotive OEMs & Tier-1 suppliers



Research labs (IITs, IISc)



Simulation-tool partners (e.g. MathWorks)



Transport & road-safety departments



Fleet & public-transport operators



Mobility startups



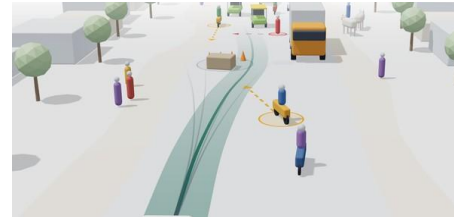
POTENTIAL IMPACTS — TARGET AUDIENCE

Potential impact on the target audience



Passengers / Drivers

Safer interaction with unpredictable road users.



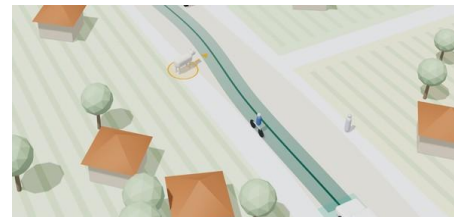
Urban Mobility

Adaptive navigation in dense mixed traffic.



Rural Mobility

Improved handling of roads without reliable lane markings.



Automotive / Mobility Industry

Foundation for Indian-road-aware autonomous systems.



Research & Academia

Platform for studying unstructured-road perception and planning.



concept illustrations



BENEFITS

Benefits of the solution (social, economic, environmental, etc.)



SAFETY

Better recognition of potential ego-path conflicts.



ADAPTABILITY

Responds to changing traffic and road conditions.



SCALABILITY

Prototype → pilot → production pathway.



INDIAN-ROAD RELEVANCE

Designed around mixed and irregular traffic behaviour.

SOCIAL · ECONOMIC · ENVIRONMENTAL

Social: safer shared roads for pedestrians, cyclists and two-wheelers.

Economic: fewer conflicts for fleets and home-grown driver-assistance capability.

Environmental: smoother, anticipatory driving with fewer abrupt stops (expected).



Business Model Canvas



The Business Model Canvas — PathSense

Designed for: PathSense	Designed by: Ctrl Alt Elite	Date: 2026	Version: Idea stage
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Key Partners - Automotive OEMs - Mobility companies - Research institutions (IITs, IISc) - Sensor / perception providers - Simulation platforms (e.g. MathWorks) - Government / transport organisations	Key Activities - Perception development - Motion prediction - Path planning - Scenario validation - Sensor integration	Value Propositions - Adaptive planning for unstructured roads - Indian mixed-traffic awareness - Safety-oriented path decisions - Explainable: every decision has a reason - Scalable autonomous-navigation architecture	Customer Relationships - Co-development pilots - Integration & tuning support - Continuous model and scenario updates	Customer Segments - Automotive manufacturers - Autonomous mobility companies - Fleet operators - Research organisations - Smart mobility programmes
	Key Resources - AI / ML models - Indian road datasets - Simulation environments - Engineering team		Channels - Direct B2B partnerships - Pilot deployments with fleets - Research collaborations - Licensable SDK / API	
Cost Structure - Engineering R&D - Data collection & annotation - Compute & simulation infrastructure - Validation & safety engineering			Revenue Streams - Software licensing to OEMs - Pilot / integration contracts - Scenario-library & validation services - Support & update subscriptions	



GOVERNMENT / NATIONAL ALIGNMENT

Connect your solution with national missions



Digital India

How PathSense aligns

Software-first, camera-based mobility intelligence built on open digital tooling.



Smart Cities Mission

How PathSense aligns

Adaptive navigation for dense, mixed urban traffic and future smart-mobility services.



Atmanirbhar Bharat

How PathSense aligns

Indigenous autonomous-driving technology designed for Indian road conditions.



Swachh Bharat / Green India

How PathSense aligns

Smoother, anticipatory driving with fewer abrupt stops — an expected efficiency benefit.



AI for India 2.0

How PathSense aligns

Applied AI solving an India-specific mobility and road-safety problem.



National Road Safety (MoRTH)

How PathSense aligns

Earlier recognition of ego-path conflicts with pedestrians, cyclists and two-wheelers.



SUPPORTING RESEARCH & SOURCES

Details / Links of the reference and research work

AUTONOMOUS DRIVING

Paden et al., “A Survey of Motion Planning and Control Techniques for Self-Driving Urban Vehicles”, IEEE T-IV, 2016

arxiv.org/abs/1604.07446



OBJECT DETECTION

Redmon et al., “You Only Look Once: Unified, Real-Time Object Detection”, CVPR, 2016

arxiv.org/abs/1506.02640



MULTI-OBJECT TRACKING

Zhang et al., “ByteTrack: Multi-Object Tracking by Associating Every Detection Box”, ECCV, 2022

arxiv.org/abs/2110.06864



MONOCULAR DEPTH

Yang et al., “Depth Anything V2”, NeurIPS, 2024

arxiv.org/abs/2406.09414



MOTION PREDICTION

Alahi et al., “Social LSTM: Human Trajectory Prediction in Crowded Spaces”, CVPR, 2016

openaccess.thecvf.com



INDIAN ROAD PERCEPTION

Varma et al., “IDD: A Dataset for Exploring Problems of Autonomous Navigation in Unconstrained Environments”, WACV, 2019

idd.insaan.iiit.ac.in



OFFICIAL SOURCE

Smart India Hackathon 2026 — Problem Statement SIH26037: Adaptive Path Planning and Collision Avoidance for Autonomous Vehicles on Unstructured Indian Roads

sih.gov.in



ROAD-SAFETY DATA & FOOTAGE

MoRTH, “Road Accidents in India 2022”; demo road videos: L. Shyamal, Wikimedia Commons (CC0)

morth.nic.in · commons.wikimedia.org